TOM SANDER

Google Scholar: https://scholar.google.com/citations?user=xrewx-sAAAAJ&hl=en
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EXPERIENCE

Meta Superintelligence Labs, FAIR - Paris
AI Research Scientist Sept 2025 – Present
PhD Research Assistant (CIFRE with École Polytechnique) March 2023 – Sept 2025

- **Provenance Release (Meta Seal, Dec. 2025):** Core contributor to Meta's official provenance offering. My open-source projects Watermark Anything and TextSeal were integrated as the foundational engines for this release.
- **Privacy Research & Engineering:** Invented TAN, now **deployed in production**. Lead author of DP-Cap (SOTA DP Image Encoder), for which I established scaling laws by orchestrating distributed training on **512+ GPUs**.
- **SOTA Research:** First author of Radioactive Watermarks (NeurIPS 2024 Spotlight), Radioactive Benchmarks, and Watermark Anything (1.1k+ stars), setting new standards for media provenance.

Meta Responsible AI Research - Paris June 2022 – Oct 2022
AI Research Scientist Intern

- **First Author ICML 2023 paper** (TAN) and achieved SOTA on ImageNet DP training within a **4-month master internship**. This method was subsequently deployed in Meta's production stack to scale privacy-preserving models.

Université de Montréal, CORAIL Lab - Montreal April 2021 – Aug 2021
M.S. Research Intern

- Co-developed **SeaPearl.jl**, an RL-based Constraint Programming solver. I designed the architecture for learning heuristics, a work recognized by the **Best Research Internship Prize** at École Polytechnique.

Université Paris-Saclay, LIMSI-CNRS - Palaiseau Sept 2020 – Jan 2021
M.S. Research Assistant

- Researched unsupervised learning methods using optimal transport for multilingual embeddings.

BNP Paribas, PACE - Paris June 2020 – Sept 2020
Data Scientist Intern (Fraud Detection)

EDUCATION

École Polytechnique (IP Paris) & Meta FAIR Dec 2022 – Sept 2025
Ph.D. in Mathematics & Machine Learning

- Thesis: *Data Tracing in Deep Learning for Privacy and Watermarking*
- Advisors: Prof. Alain Durmus (CMAP) & Chuan Guo (Meta FAIR)

École Normale Supérieure Paris-Saclay (MVA) 2021 – 2022
M.S. in Computer Vision and Machine Learning

- Highest Honors (Mention Très Bien). GPA: 4.0 (18.3/20), Top 5%.

École Polytechnique (IP Paris) 2018 – 2022
Ingénieur Polytechnicien Program (B.S. & M.S.)

- Admitted via highly selective nationwide competitive exam ("Concours X", **top 1%** acceptance rate).
- Major in Applied Mathematics and Computer Science. GPA: 3.96/4.0.

Lycée Henri-IV - Paris 2016 – 2018
Classes Préparatoires aux Grandes Écoles (MPSI / MP)*

- Intensive bachelor-level program in Advanced Mathematics & Physics with Computer Science Option.

FIRST AUTHOR PUBLICATIONS

How Good is Post-Hoc Watermarking With Language Model Rephrasing? *Preprint*, 2026
Pierre Fernandez*, Tom Sander* et al.

Watermark Anything with Localized Messages *ICLR*, 2025
Tom Sander*, Pierre Fernandez*, Alain Durmus, Teddy Furron, Matthijs Douze

Rethinking the Role of Verbatim Memorization in LLM Privacy *NeurIPS*, 2025
Tom Sander, Bargav Jayaraman, Mark Ibrahim, Chuan Guo, Kamalika Chaudhuri

Detecting Benchmark Contamination through Watermarking *ICLR Workshop*, 2025
Tom Sander, Pierre Fernandez, Saeed Mahloujifar, Alain Durmus, Chuan Guo

Watermarking Makes Language Models Radioactive ***NeurIPS (Spotlight)***, 2024
Tom Sander*, Pierre Fernandez*, Alain Durmus, Matthijs Douze, Teddy Furron

Differentially Private Representation Learning via Image Captioning (DP-Cap) *ICML*, 2024
Tom Sander*, Yaodong Yu*, Maziar Sanjabi, Yi Ma, Alain Durmus, Kamalika Chaudhuri, Chuan Guo

Implicit Bias of Noisy-SGD: With Applications to DP Training *AISTATS*, 2024
Tom Sander, Maxime Sylvestre, Alain Durmus

TAN Without a Burn: Scaling Laws of DP-SGD *ICML*, 2023
Tom Sander, Pierre Stock, Alexandre Sablayrolles

** denotes equal contribution – Full publication list available on my Google Scholar*

CODING SKILLS & OPEN SOURCE

- **Deep Learning:** PyTorch (Expert; Distributed training, FSDP), TensorFlow, JAX.
- **Languages:** Python, Julia (contributor to `SeaPearl.jl`), C++, OCaml, Java.
- `facebookresearch` **Open Source Leadership:**
  - **Watermark Anything Models (`wam`):** Co-lead, **1.1k+ stars**.
  - **Lead Developer:** `radioactive-watermark`, `dp-cap`, `textseal`, `tan`.

SELECTED INVITED TALKS

Séminaire Palaisien - Palaiseau: *Watermarking for Active Content Tracing* 2025
NeurIPS in Paris (Oral) - Paris: *Watermarking Makes Language Models Radioactive* 2024
NeurIPS Panel - Vancouver: *Panel on Watermarking led by Tom Goldstein* 2024
Meta GenAI Safety: *Beyond Verbatim Memorization in LLMs* 2024
ENS Paris (DMA): *Watermarking Makes Language Models Radioactive* 2024
Mila (Montreal): *TAN without a Burn: Scaling Laws of DP-SGD* 2022

AWARDS

Spotlight at NeurIPS: *Top 10% of accepted papers (Radioactive Watermarks)* 2024
Best Research Internship, École Polytechnique 2022
Outstanding Investment Award, École Polytechnique (VP Entrepreneurship, raised $10k) 2022
First Prize in Mathematics, École Alsacienne (High School): *Best student of the year* 2016

SERVICE & INFO

Reviewer: NeurIPS (2023-2024), ICML (2023), ICLR (2025), CVPR/ICCV (2025), JMLR.

Teaching: Mathematics Lecturer at USTH (Hanoi, Vietnam) during 6-month Civil Service (2019).

Mentoring: Scientific Advisor for Jimini.ai (Station F 2023 program).

Languages: English (Fluent), French (Native), Chinese (Intermediate), Hebrew (Intermediate).